

# Realtime-Rendering of Dynamic Scenes with Neural Radiance Fields

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Figure 1: Two views of a test sequence showcasing different perspectives of the volumetric scene and two closeups that show the visual fidelity. Our method is able to render the sequence in real-time from novel viewpoints on a Desktop computer.

## ABSTRACT

Digitizing the humans and scenes around us is a challenging research topic, with results finding new and more widespread implementation in modern applications. This paper addresses the challenge of real-time rendering of dynamic 3D scenes. The proposed approach utilizes Instant Neural Graphics Primitives (iNGP), training a radiance field for each individual frame. To achieve playback rates suitable for interactive VR experiences, we developed efficient data storage and loading methods. We present a modular framework for playing real-time free-viewpoint videos on a conventional Desktop PC, to showcase research advancements of dynamic radiance fields.

**Index Terms:** NeRF, Volumetric Video, Virtual Humans.

## 1 INTRODUCTION

The digital portrayal of objects and environments has become indispensable across diverse applications, including virtual reality (VR), gaming, scientific visualization, and cultural preservation. Achieving photorealistic and interactive visualizations of 3D scenes, however, remains a challenge due to the computational demands of accurately modeling and rendering complex geometry and lighting. One significant advancement in this field is the development of dynamic 3D sequences, sometimes referred to as volumetric video [3]. Traditionally, these sequences have been represented using polygonal meshes, which, however, struggle to capture intricate visual effects such as complex lighting and transparency. Neural Radiance Fields (NeRF) [1] have recently revolutionized the visualization of static scenes by using machine learning to synthesize highly realistic views from a sparse set of input images. Despite their groundbreaking results, traditional NeRF implementations are computationally highly intensive and thus have limitations in their applicability to dynamic, real-time scenarios.

We address these limitations by investigating how Instant Neural Graphics Primitives (iNGP) [2] can be leveraged for real-time rendering of dynamic 3D content. This work introduces a simple, modular framework capable of achieving competitive results without compromising rendering speed or quality. The proposed method involves training independent static NeRFs for each frame in a video sequence, combined with an optimized data storage and loading pipeline to enable real-time playback.

## 2 METHOD

We developed a strategy to enable seamless and interactive playback of neural volumetric videos, mimicking conventional video playback for easy integration. This involves loading a new static NeRF for each frame in order to render the volumetric video. For that, we employ a multi-view camera setup to capture diverse viewpoints, ensuring a comprehensive scene representation.



Figure 2: Storing and loading the density grid, here displayed in slices, is required to achieve fast rendering speeds.

In order to achieve real-time playback, we use the significantly faster implementation of Instant Neural Graphics Primitive (Instant-NGP) as the basis for the radiance field in this work. Instant-NGP employs three core components: a hash table that maps 3D coordinates to feature vectors for MLP processing, a density grid that identifies regions with nonzero density for ray tracing (Figure 2), and the MLP layers that are repeatedly evaluated for each ray to synthesize frames. Those components can be stored as various high-dynamic-range images (HDR images) in EXR format in order to be able to save and load 32-bit float values without any loss of quality. The playback process involves strategically replacing the hash table, the density grid, and the MLP layers in each frame. To enhance efficiency, components are optimized through

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techniques such as thresholding, parallelization, and quantization, significantly reducing computational demands per frame.

To further improve performance and quality, we pre-process the training data to address specific challenges in dynamic scenes. For example, certain regions in the scene, such as object-background transitions, can introduce artifacts that disrupt seamless rendering and are challenging to remove without creating unnatural edges. A custom erosion mask was created to exclude such problematic areas during training. This process generates a binary representation of the data, modifies it to isolate the object-background transition, and excludes those regions from training. This design ensures that the network interprets these areas as undefined rather than empty, preserving the integrity of the rendering while avoiding the introduction of artifacts.

Overall, the methodology was guided by the aim to balance efficiency, adaptability, and visual fidelity, ensuring that neural volumetric video playback can be achieved interactively and in real time.

### 3 EVALUATION

To evaluate the framework, two key aspects are examined: rendering performance and visual quality. Rendering performance focused on frame rates and memory usage, particularly during sequential frame loading and real-time rendering. Visual quality was evaluated through subjective assessments and quantitative metrics such as structural similarity index (SSIM) and peak signal-to-noise ratio (PSNR). Experiments were conducted on five different base sizes (huge, big, medium, small, tiny) to optimize the framework's performance. Baseline tests established initial metrics for frame rate, memory usage, and render quality. Data storage methods, including floating-point arrays and binary masks, were compared, and the use of run-length encoding for density grid compression was explored. Thresholding was applied to eliminate free-floating artifacts and reduce memory usage, analyzing its impact on visual quality and efficiency. Temporal consistency was improved by addressing flickering and using erosion masks. Multithreading was implemented to accelerate data loading, comparing strategies where threads either loaded separate datasets or collaborated on single frames. Finally, quantization data compression was employed to reduce hash table size while preserving rendering accuracy. To improve performance, caching was utilized to store recently accessed data in faster memory for quicker retrieval. After the initial load, hash table and density grid data are cached, nearly doubling the frame rate for most sequences. However, sequences with larger hash tables and higher per-frame data volumes see less dramatic improvements. Further reduction in data per frame and overhead was achieved by combining MLP layers and hash table into a single file as well as reducing the density grid from a float array to a binary representation. For the medium size with  $2^{16}$  feature vectors, the video surpasses 25 fps - the acquisition frame rate of our scenes - only after caching the data (Figure 3). Through the optimization of combining MLP layers with the binary density grid representation, the playback achieves 35 frames per second (fps) when uncached, and 98 fps once cached, with an average peak signal-to-noise ratio (PSNR) of 23.7 across all camera views. The motion of the virtual humans is shown at the recorded rate of 25 fps. With the smaller configurations, we are able to synthesize multiple novel views per single frame of human pose. This allows for a smoother playback experience suitable for VR headsets, which require at least 60-90 fps.

Our methodology achieves comparable visual quality to those documented in iNGP for each individual frame. The size of the hash table has the biggest influence on both playback speed, achieved rendering quality and storage requirements. Choosing the smallest version allows one to save a one minute long sequence in only 3.4 GB instead of the 569 GB of the original training data. The

largest configuration requires 38 GB for the entire video. Another promising technique is post-training quantization of the hash table. Experimental results showed that data could be quantized to multiples of  $\frac{1}{8}$  without losing any visual fidelity.

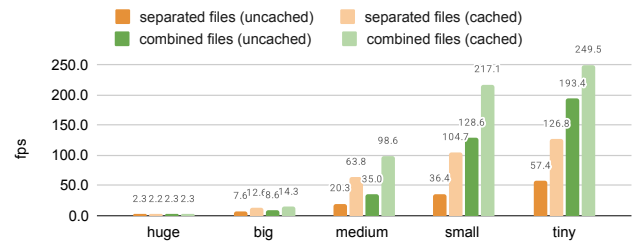


Figure 3: Achieved playback rate in frames per second. Combining the hash table and MLP layers into a single file in combination with a binary density grid yields an additional performance boost per frame.

### 4 CONCLUSION

This work demonstrates the feasibility of creating real-time free-viewpoint video using a modular framework based on a standard NeRF implementation. By leveraging the efficiency of Instant Neural Graphics Primitives (iNGP) and introducing targeted optimizations in data storage and retrieval, the system achieves high-quality rendering of dynamic scenes, including human subjects, without the need for more complex neural networks or increased per-frame render times. Originally tailored for rendering humans against a transparent backdrop, the framework's optimizations are applicable to broader scenes, excluding density grid and erosion mask simplifications. The approach of training independent static NeRFs for each frame not only simplifies the pipeline but also ensures flexibility for future enhancements. Although successful, the framework has constraints, such as occasional temporal artifacts like flickering, which highlight the need for more sophisticated solutions to ensure temporal consistency. Future work could address this by incorporating time as an additional parameter in the neural network or by using prior frame data to improve initialization and training efficiency. Additionally, employing advanced compression techniques like quantization-aware training could further optimize the hash table's storage requirements, enabling even more efficient data handling. This study establishes a robust basis for future investigations, demonstrating that streamlined modular neural rendering methods can deliver high-quality, real-time outcomes while maintaining accessibility and adaptability.

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